

Technology Stack

Date	25September2026	MaximumMarks	2Marks
TeamID	SWTID-2026-7796	TeamLeader	PoornaLakshmiM
Project Name	StreamliningITProcurement:AutomatingStandardLaptopOrderswithFlowDesigner		

TechnologyStack Details

Sn	ArchAitectureComponent/ Layer	TechnologyChosen	Justification/Purpose
1	Frontend/Client-Side (e.g., React, Angular,HTML/CSS)	ServiceNowServiceCatalog ServicePortal/NextExperienceUI	Provides standard, responsive self-service forms forrequisitioningstandardlaptops(e.g., Lenovo Carbon X1).Gives end-users real-timevisibilityintotechnical specifications, pricing, optional add-ons,anddynamicdeliverystagetracking.
2	Backend/Server-Side (e.g.,Node.js,Python,Java)	ServiceNowFlowDesigner Workflow Studio &JavaScriptEngine	Deliversmodern,low-codeworkflowautomationtriggerednativelybyServiceCatalogsubmissions.Executes business logic to monitor approvalstates and automatically instantiate and routefulfillmenttaskstotheHardwaregroupasSystemUser.
3	Database /DataStorage (e.g., MySQL, MongoDB,PostgreSQL)	ServiceNowRelationalDB (MariaDB/MySQLMulti-instanceSchema)	Stores all transactional tables seamlessly linked throughreferences:ServiceCatalogitems (sc_cat_item),Requests(sc_request), RequestedItems(sc_req_item),Approvals (sysapproval_approver),andTasks(sc_task).

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Technology Stack Details—(Continued)

S.No	Architecture Component/ Layer	Technology Chosen	Justification/Purpose
4	Cloud/Hosting/ Deployment (e.g., AWS, Azure, Docker)	ServiceNow Cloud(PaaS) Enterprise Multi-Instance Architecture	Provides high-availability, enterprise-grade cloud hosting(dev instance dev409451/dev429721) with automatic failover, managed database clustering, built-in system security patches, and instant zero-downtime flow activation.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	ServiceNow Update Sets/ Git ServiceNow Studio Source Control	Enable team collaboration across 4 developers (Poorna Lakshmi M, Vajitha T, Sneha N, Kanisiya G). Tracks changes across flows, catalog item definitions, and process engine linkages for clean staging and promotion.
6	Third-Party APIs/Other Tools (e.g., Stripe, Twilio, Google Maps)	ServiceNow REST API & Spoke Process Automation/Table API	Expose stable endpoints and outbound notification hooks to inform users and hardware configuration agents. Integrates with email notification daemons and allows integration into corporate IT asset management systems.